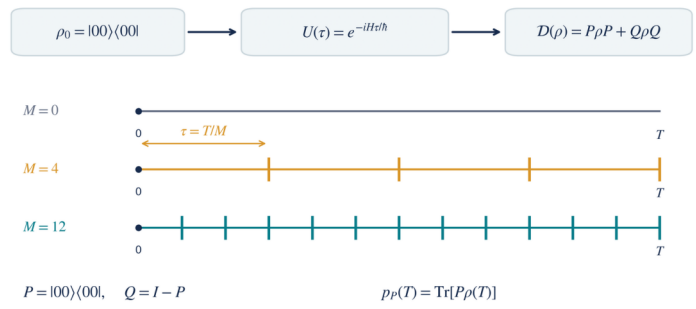
Quantum Zeno effect · figure selection sheet

01 Measurement protocol · wide

01 Measurement protocol

Ideal instantaneous binary projections; outcomes discarded unless explicitly post-selected.

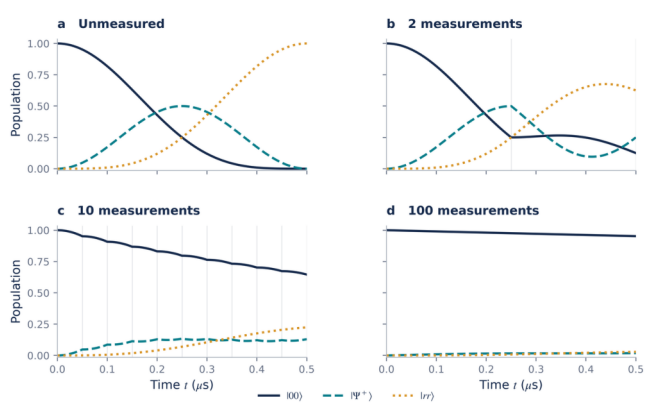


Non-selective: sum all outcomes. All-success: $S_M = |\langle 00|U(T/M)|00\rangle|^{2M}$. Events include T.

02 Population evolution · wide

02 Population evolution

$\Omega/2\pi = 1$ MHz · $\Delta = \phi = 0$ · $\rho_0 = |00\rangle\langle 00|$ · $V = 0$

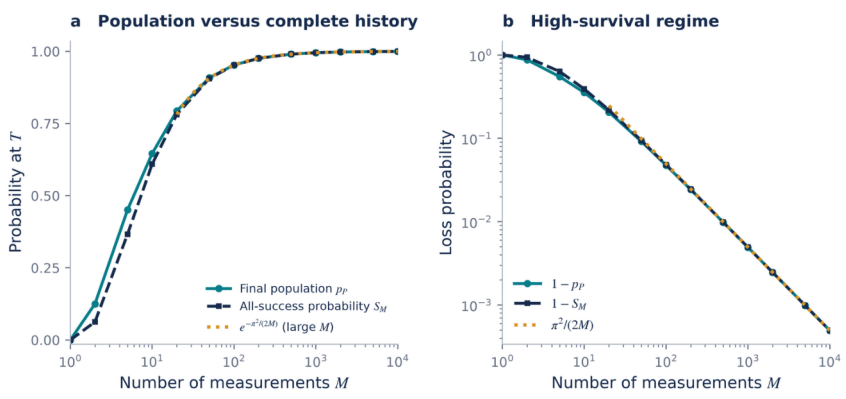


$|\Psi^+\rangle = (|0r\rangle + |r0\rangle)/\sqrt{2}$; the antisymmetric state remains dark. Non-selective measurement.

03 Zeno survival curve · wide

03 Frequent measurements suppress transitions

$V = 0 \cdot T = 0.5\,\mu$s · $\Omega/2\pi = 1$ MHz

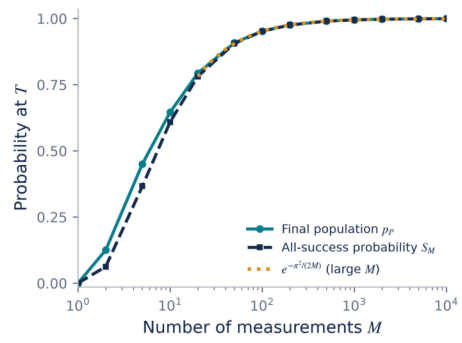


Non-selective population includes return histories; S_M excludes them. $M = 0$: $p_P \simeq 0$. $\Delta = \phi = 0$; initial $|00\rangle$.

03 Zeno survival curve · single

03 Zeno survival

$V = 0 \cdot T = 0.5\,\mu$s · $\Omega/2\pi = 1$ MHz

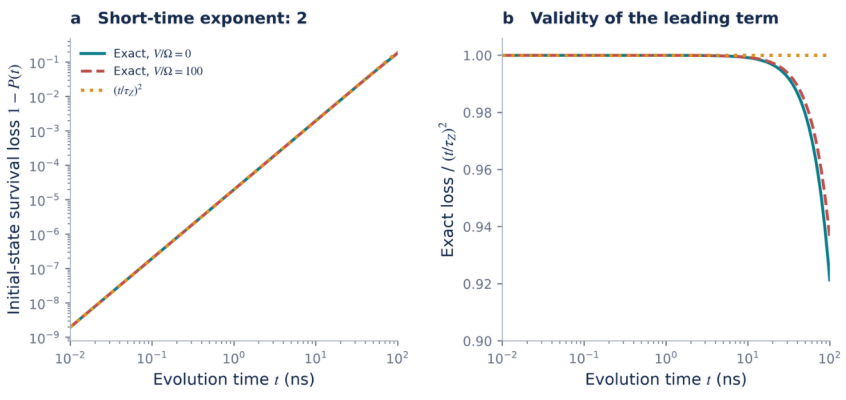


Non-selective population includes return histories.
$M = 0$: $p_P = 0$; $\Delta = \phi = 0$; initial $|00\rangle$.

04 Short-time quadratic law · wide

04 The quadratic origin of Zeno suppression

$\Omega/2\pi = 1$ MHz · $\Delta = \phi = 0$ · $\rho_0 = |00\rangle\langle 00|$

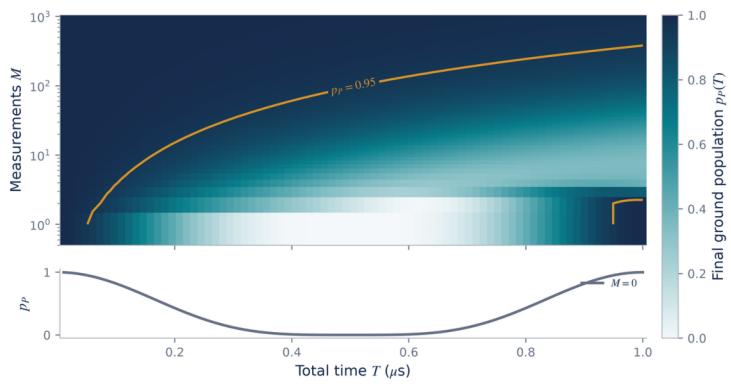


$\tau_Z = \sqrt{2}/|\Omega| = 225.08$ ns for every V. The quadratic window depends on higher Hamiltonian moments.

05 Zeno regime heatmap · wide

05 Where transitions are suppressed

$\Omega/2\pi = 1$ MHz · $\Delta = \phi = 0$ · $\rho_0 = |00\rangle\langle 00|$ · $V = 0$

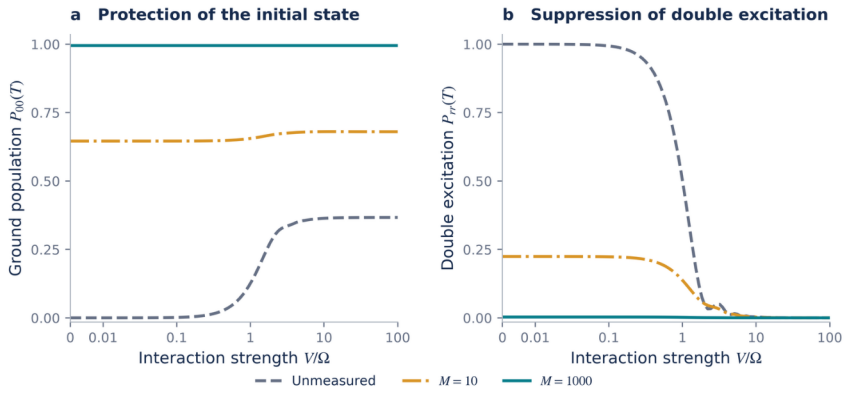


The unmeasured control exposes coherent revivals. A high endpoint population alone does not establish Zeno suppression.

06 Interaction dependence · wide

06 Measurement inhibition and interaction blockade

$\Omega/2\pi = 1$ MHz · $\Delta = \phi = 0$ · $\rho_0 = |00\rangle\langle 00|$ · $T = 0.5\,\mu$s

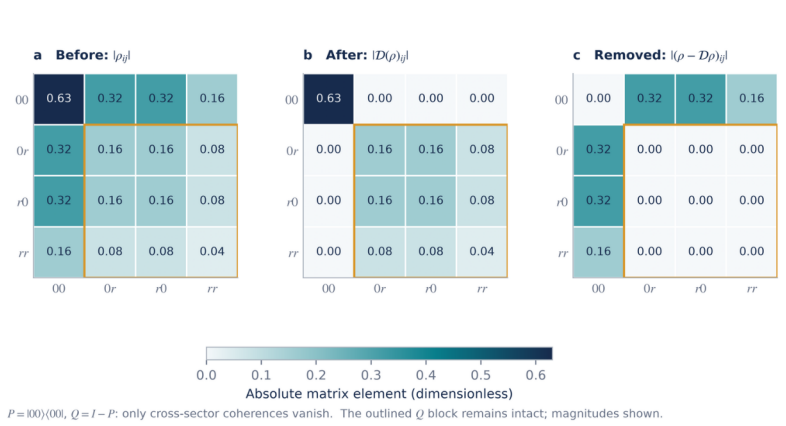


Strong V inhibits double excitation without freezing $|00\rangle$. Matched $M = 0$ controls separate the mechanisms.

07 Density-matrix coherence · wide

07 What a non-selective measurement removes

$\Omega/2\pi = 1$ MHz · $\Delta = \phi = 0$ · $\rho_0 = |00\rangle\langle 00|$ · $V = 0$ · $t = 0.15\,\mu$s

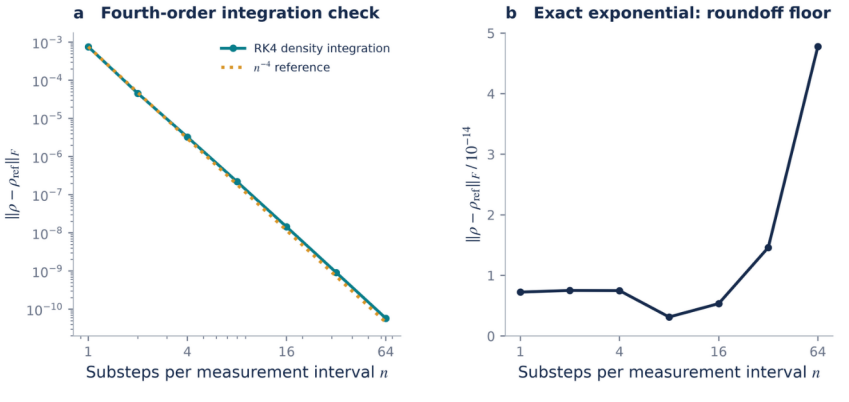


$P = |00\rangle\langle 00|$, $Q = I - P$: only cross-sector coherences vanish. The outlined Q block remains intact; magnitudes shown.

08 Numerical convergence · wide

08 Numerical accuracy and convergence

$\Omega/2\pi = 1$ MHz · $\Delta = \phi = 0$ · $\rho_0 = |00\rangle\langle 00|$ · $V/\Omega = 1$ · $M = 8$ · $T = 0.5\,\mu$s

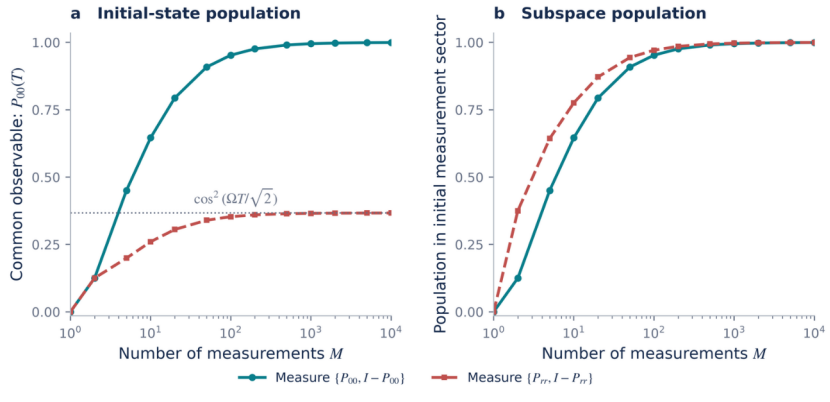


Reference: independent spectral unitary + channel powers. Production uses exact static propagation; no integration timestep is fitted.

09 Measurement-subspace comparison · wide

09 The measured subspace determines the dynamics

$\Omega/2\pi = 1$ MHz · $\Delta = \phi = 0$ · $\rho_0 = |00\rangle\langle 00|$ · $V = 0$ · $T = 0.5\,\mu$s



Right: rank-one P_{00} versus rank-three $I - P_{rr}$. Reversing binary outcome labels leaves the non-selective channel unchanged.